

Supplementary Material for: The Loyola Index Family: What a Degree-Imbalance Coordinate Can and Cannot Resolve

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This supplement accompanies the paper “The Loyola Index Family: What a Degree-Imbalance Coordinate Can and Cannot Resolve”. Section, table, figure, equation and proposition numbers without the prefix S refer to the main paper. Data, scripts and the verifier are available at <https://github.com/Singati2/LOYOLA-PAPER>.

S1 Octane property data: provenance audit and correlation heatmap

Table 4 of the main paper reports the Pearson correlation r between each index and each of the five physicochemical properties of the 18 octane isomers (boiling point T_B , enthalpy of formation ΔH_f , enthalpy of vaporization ΔH_{vap} , entropy S , acentric factor ω ; property data traced to the NIST WebBook [5] (the provenance file records, for each value, the Web-

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Book entry or, where the WebBook lists no value, the compilation consulted), except the acentric factors, which NIST does not list and which were taken from KDB-derived compilations; the same five properties form the octane benchmark used in [6]). Units are: T_B in $^{\circ}\text{C}$; ΔH_f and ΔH_{vap} in kcal mol^{-1} (the latter at 298 K); S in $\text{cal mol}^{-1}\text{K}^{-1}$; and ω dimensionless. Values for ΔH_{vap} and entropy S vary slightly across compilations; all comparisons here are made within this single, internally consistent dataset. Calculations use this author-compiled property table attributed to the NIST WebBook. All 90 molecule-level values were traced against authoritative sources (NIST WebBook first; KDB-derived compilations for the acentric factor, which NIST does not list), and classified under a strict taxonomy in which “verified” requires agreement at the value’s reported precision (a converted value exactly half a display unit away counts as agreement at reported precision): 3 match exactly, 29 match a specific recorded determination at reported precision after unit conversion (gas phase confirmed for formation enthalpies and entropies), 40 agree within the audit tolerances, the source’s stated uncertainty or the span of several determinations without matching any one of them at reported precision, 10 lie within explained variation among authoritative determinations (1947 API Project 44 entries superseded by later TRC revisions), 7 could not be verified against a reachable authoritative value (six gas-phase entropies absent from the NIST WebBook, for which the 1947 API Project 44 tables give values differing by 0.07–2.2 units, as they do for isomers whose later revisions were confirmed, and the 298 K vaporization enthalpy of 2,2,3,3-tetramethylbutane: this compound is a solid at 298 K (triple point 374 K); the NIST WebBook lists standard vaporization enthalpies of 42.94 and 42.91 kJ mol^{-1} (about 10.26 kcal mol^{-1}) together with sublimation enthalpies, whereas the benchmark value used here, 8.41 kcal mol^{-1} (35.19 kJ mol^{-1}), is that of the standard octane benchmark compilation and matches no listed determination, so its physical reference state is unresolved; Section S3 reports the ΔH_{vap} analysis with this isomer excluded), and one value is a marginal conflict: the acentric factor of 2,2,3,3-tetramethylbutane (0.247 vs. compiled 0.251); the acentric factor of 2,2,4-trimethylpentane (0.305 vs. compiled 0.303) agrees only within tolerance.

Substituting the alternative compiled acentric factors changes several of the ω correlations in Table 4 of the main paper at the third decimal (e.g. M_2 moves from -0.988 to -0.986) but does not change the property-wise winning descriptor or any substantive interpretation; the full sensitivity computation and the machine-readable provenance trace accompany the reproducibility package. For 3,3-dimethylhexane, 3-ethyl-3-methylpentane, 2,2,3-trimethylpentane and 2,3,3-trimethylpentane, values of ΔH_{vap} that differ from the NIST WebBook by $0.05\text{--}0.13\text{ kcal mol}^{-1}$ circulate in octane benchmark tables; we use values consistent with NIST (Appendix A of the main paper): two exact conversions of NIST determinations (8.97, 9.08) and two values that lie within the span or stated uncertainty of the NIST determinations (8.83, which also equals the API Project 44 value at the displayed precision, and 8.90, which equals the compiled octane dataset value 8.897 at the displayed precision). Six gas-phase entropies could not be verified against a primary source (see above); four entropies (*n*-octane, 2,2-dimethylhexane, 2,2,4- and 2,3,3-trimethylpentane) differ from those of a widely used octane QSPR compilation by $0.04\text{--}2.29$ units (the 2,3,3-trimethylpentane value 102.06 is a compiled-dataset value; see the data-quality appendix of the main paper); substituting the compilation values changes all twelve S correlations in Table 4 of the main paper (by up to 0.043 in $|r|$) but not the winning descriptor (HM). Apart from the four corrected vaporization enthalpies and the corrected transcription of the 2,3,3-trimethylpentane entropy (data-quality appendix of the main paper, item (ii)), the benchmark values are preserved as supplied for comparability.

Correlation details for Table 4 of the main paper. The best correlation per property is split among the reductions and the pure- γ points: the modified second Zagreb index $LO(-1,0,0)$ is strongest on T_B and ΔH_f , the hyper-Zagreb index $LO(0,2,0)$ on S , the second Zagreb index $LO(1,0,0)$ on ω , and the fixed pure- γ benchmark point $LO(0,0,1)$ on ΔH_{vap} ($|r| = 0.984$, with $LO(0,0,2)$ second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the corresponding classical indices to floating-point precision, as they must.

For each property, a descriptive post-selection paired bootstrap interval for the observed top-two $|r|$ difference ($B = 10\,000$, resampling the 18 isomers jointly, `default_rng(1)`) included zero; these intervals were not multiplicity-adjusted across the five properties and do not establish equality or equivalence. Individual bootstrap half-widths range from 0.016 to 0.096 across the headline cells (percentile bootstrap, $B = 10\,000$, `default_rng(0)`). The value of $LO(0,0,1)$ on ΔH_{vap} is a fixed benchmark point, not a tuned result. Recomputing each bolded correlation with single isomers deleted shifts $|r|$ by at most 0.027, indicating that no single isomer drives the leading correlations.

Descriptor redundancy. The redundancy is in fact systemic, and it caps how much meaning any small $|r|$ difference in Table 4 of the main paper can carry. Across the twelve compared descriptors on the 18 isomers, fourteen descriptor pairs have $|r| \geq 0.99$ (the extreme being R versus $H/2$ at $r = 0.9997$), and a principal component analysis of the standardized 18×12 descriptor matrix places 90.3% of the variance on the first component and 99.05% on the first two, with an effective rank (participation ratio of the eigenvalue spectrum) of 1.2. The twelve-index comparison set is thus essentially a two-dimensional descriptor space, and differences in the third decimal of $|r|$ between near-collinear descriptors should not be read as meaningful rankings. Figure S1 shows the signed- r heatmap of the compared indices against the five properties; it displays the values of Table 4 of the main paper.

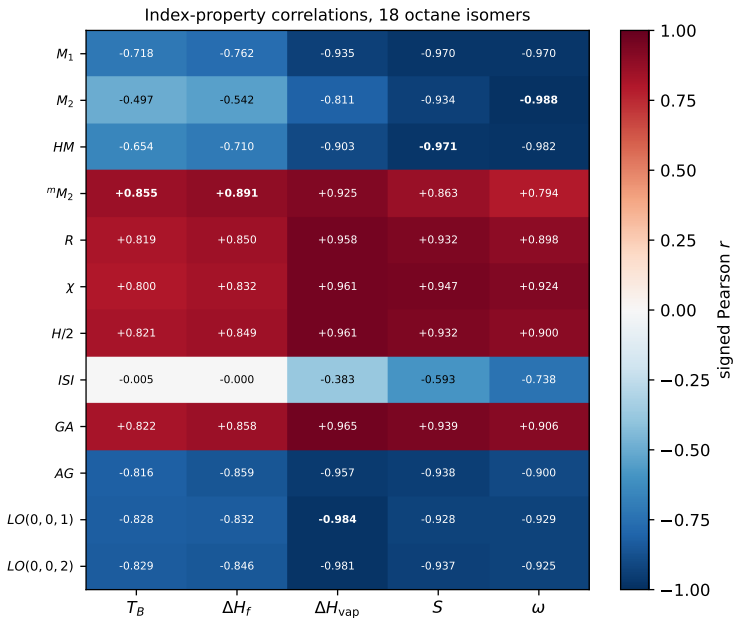


Figure S1. Signed Pearson r between the Loyola reductions, the fixed pure- γ benchmark points and the five octane physicochemical properties on the 18 isomers.

S2 Effect of parameter tuning near the best indices

The benchmark of Section 6.1 of the main paper tunes no parameters: it compares only the exact reductions and the two fixed pure- γ benchmark points. It is nonetheless natural to ask whether small, data-driven adjustments of (α, β, γ) near the best-correlating indices could improve the fit. We probe this directly as an exploratory check. For each of the four anchors carrying the strongest column correlations in Table 4 of the main paper — $M_2 = LO(1, 0, 0)$, $HM = LO(0, 2, 0)$, ${}^mM_2 = LO(-1, 0, 0)$, and the pure- γ point $LO(0, 0, 1)$ — we draw 50 random parameter triples within ± 0.3 (to two decimal places) of the anchor and correlate each against the five octane properties. The four pools are drawn, in the order listed,

from one pseudo-random stream (NumPy `default_rng(42)`; each coordinate offset is a uniform draw on $[-0.3, 0.3]$ rounded to two decimals); the percentile bootstrap uses `default_rng(0)`. Table S1 reports, for every anchor–property cell, the anchor correlation $|r|_0$, the best of the 50 perturbations in sample $|r|_{50}$, the in-sample gain Δ_{in} , the percentile-bootstrap half-width ($B = 10\,000$), the leave-one-out correlation of the anchor LOO_0 , and a *nested* leave-one-out correlation LOO_{50} in which, within each of the four globally selected anchor pools, the best of the 50 candidates is selected using the 17 training isomers of each fold only and evaluated on the held-out one, with the resulting out-of-sample gain Δ_{out} . The anchors themselves were chosen from the complete-data correlations of Table 4 of the main paper, and the 18 pooled fold predictions share heavily overlapping training sets, so these values are descriptive rather than independent replications.

Table S1. Decimal parameter tuning near the four best-correlating anchors on the 18 octane isomers (50 random triples within ± 0.3 of each anchor). Columns: anchor correlation $|r|_0$; best-of-50 in sample $|r|_{50}$; in-sample gain Δ_{in} ; percentile-bootstrap half-width $\pm\text{boot}$ ($B = 10000$); leave-one-out correlation of the anchor LOO_0 ; foldwise nested leave-one-out correlation of the tuned family, computed within the globally selected anchor pool, LOO_{50} ; out-of-sample gain $\Delta_{\text{out}} = \text{LOO}_{50} - \text{LOO}_0$. Gains are computed from full-precision values and rounded once, so a displayed gain may differ by one unit in the last place from the difference of the displayed operands.

Property	$ r _0$	$ r _{50}$	Δ_{in}	$\pm\text{boot}$	LOO_0	LOO_{50}	Δ_{out}
<i>near</i> $M_2 = \text{LO}(1, 0, 0)$							
T_B	0.497	0.598	+0.100	0.353	0.331	0.458	+0.127
ΔH_f	0.542	0.635	+0.093	0.358	0.395	0.540	+0.145
ΔH_{vap}	0.811	0.876	+0.066	0.185	0.768	0.838	+0.071
S	0.934	0.955	+0.021	0.073	0.909	0.935	+0.025
ω	0.988	0.995	+0.007	0.018	0.985	0.985	0.000
<i>near</i> $HM = \text{LO}(0, 2, 0)$							
T_B	0.654	0.740	+0.085	0.257	0.545	0.647	+0.102
ΔH_f	0.710	0.786	+0.076	0.255	0.645	0.741	+0.097
ΔH_{vap}	0.903	0.944	+0.041	0.082	0.871	0.921	+0.049
S	0.971	0.973	+0.002	0.030	0.961	0.957	-0.004
ω	0.982	0.989	+0.007	0.025	0.979	0.982	+0.003
<i>near</i> ${}^mM_2 = \text{LO}(-1, 0, 0)$							
T_B	0.855	0.864	+0.009	0.096	0.781	0.759	-0.021
ΔH_f	0.891	0.897	+0.006	0.087	0.865	0.837	-0.028
ΔH_{vap}	0.925	0.966	+0.041	0.068	0.900	0.953	+0.053
S	0.863	0.917	+0.054	0.192	0.810	0.878	+0.068
ω	0.794	0.881	+0.088	0.246	0.730	0.853	+0.123
<i>near</i> $\text{LO}(0, 0, 1)$							
T_B	0.828	0.835	+0.007	0.161	0.771	0.764	-0.007
ΔH_f	0.832	0.842	+0.010	0.152	0.800	0.807	+0.006
ΔH_{vap}	0.984	0.985	+0.001	0.016	0.978	0.978	-0.001
S	0.928	0.945	+0.017	0.046	0.887	0.915	+0.028
ω	0.929	0.945	+0.016	0.074	0.907	0.926	+0.019

Three patterns emerge. First, every in-sample gain Δ_{in} is smaller than the corresponding single-descriptor bootstrap uncertainty scale; this comparison is descriptive and is not a formal paired or post-selection test of the gain. Second, under the foldwise nested evaluation (within the globally selected anchor pools) these descriptive point-estimate gains do not vanish — tuning near M_2 raises the out-of-sample correlation for T_B and ΔH_f by +0.13 and +0.15, and tuning near mM_2 raises ω by +0.12 — but in every such case the anchor was a *weak* descriptor for that property, and the tuned correlation only climbs part way toward the level a strong reduction already attains (tuning near M_2 lifts T_B to 0.46, whereas mM_2

already reaches 0.78). Where the anchor is already the column winner — M_2 on ω , HM on S , $LO(0,0,1)$ on ΔH_{vap} — the nested gain for this pool seed is -0.0003 , -0.004 and -0.001 . These near-zero values are specific to the seed: re-drawing the pools with seeds 0–99, the nested gain for M_2 on ω is small but positive in most seeds (median $+0.003$, positive in 97 of 100 seeds; repository script `tuning_pool_seed_sweep.py`), so the defensible statement is that tuning a column winner changes its nested correlation by at most a few thousandths. Third, tuning near the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ does not exceed the best fixed index under nested leave-one-out on any of the five properties (Table S2). On ΔH_{vap} , where the best fixed index is the pure- γ point $LO(0,0,1)$ itself, the tuned value exceeds it by 0.001 in sample (0.985 vs 0.984) and falls 0.001 below it under leave-one-out (both 0.978 at three decimals; differences of full-precision values); these post-selection differences are negligible. On T_B , ΔH_f , S , and ω the best fixed reduction is higher both in and out of sample.

Table S2. Decimal tuning near the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ versus the best fixed index in the tested comparison set for each octane property. The best fixed index is ranked by in-sample $|r|$ (its leave-one-out value in parentheses); the decimal columns give the best over two pools of 50 random triples within ± 0.3 of $LO(0,0,1)$ and of $LO(0,0,2)$, in sample and under foldwise nested leave-one-out within the globally selected pools; the $LO(0,0,1)$ pool is exactly the pool of Table S1, and the $LO(0,0,2)$ pool is the next 50 triples of the same `default_rng(42)` stream. Indices are ranked by in-sample $|r|$ rather than LOO, as the latter is inflated by degenerate fits for near-zero-correlation indices. Tuning does not exceed the best fixed index under LOO on any property; in sample it exceeds it only on ΔH_{vap} , by 0.001, a negligible post-selection difference that does not establish superiority.

Property	best fixed	$ r _{\text{fix}}$	LOO _{fix}	$ r _{\text{dec}}$	LOO _{dec}
T_B	mM_2	0.855	0.781	0.835	0.764
ΔH_f	mM_2	0.891	0.865	0.850	0.815
ΔH_{vap}	$LO(0,0,1)$	0.984	0.978	0.985	0.978
S	HM	0.971	0.961	0.945	0.915
ω	M_2	0.988	0.985	0.945	0.926

In short, local tuning raised several nested leave-one-out point esti-

mates for weak anchors, and except for the 0.001 in-sample ΔH_{vap} margin noted above did not exceed the best fixed benchmark in the tested comparison set; no formal superiority claim is made. All tabled nested values correspond to positive signed prediction–observation correlations with positive predictive Q^2 , and a constant-descriptor control confirms that absolute pooled correlation alone can be inflated for degenerate descriptors (cf. the caveat in the caption of Table S2).

S3 Matched-budget comparison in the search box $(-2, 2)$

The analyses above tune around fixed anchors and are descriptive. The sharper question raised by the Gutman–Milovanović framing is direct: given identical search budgets, does freeing γ improve out-of-sample prediction beyond an optimized parent $M_{\alpha,\beta}$? We answer it with a fully nested leave-one-out design. For each property and each outer fold (one isomer held out), 200 candidate parameter vectors are drawn once from $\text{Uniform}(-2, 2)$ per coordinate (triples (α, β, γ) for the Loyola model, pairs (α, β) with $\gamma = 0$ for the parent; identical budget; candidates rounded to three decimals, drawn from NumPy `default_rng(12345)` for Table S3, Loyola triples first); on the 17 training isomers only, every candidate is scored by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression, the best candidate is selected, the regression is refit on all 17, and the held-out isomer is predicted. No quantity computed on the held-out isomer enters selection or fitting at any point.

The short answer is no: once the parent is searched over a box that does not truncate it, freeing γ gives no consistent gain (Table 5 and Section 6.2 of the main paper). We first report the comparison in the box $(-2, 2)$, because it shows how the apparent gain arose. In that box the answer is mixed (Table S3): for the illustrative seed, freeing γ improves nested predictive performance on two of the five properties (T_B , $\Delta Q^2 = +0.136$, and ΔH_{vap} , $\Delta Q^2 = +0.086$) and degrades it on three (ΔH_f , S , ω), with the ΔH_f degradation driven by selection instability: the parent model se-

Table S3. Matched-budget nested leave-one-out comparison of the optimized parent $M_{\alpha,\beta}$ (GM) and the optimized Loyola family (γ free) on the five octane properties. Panel (a) is a *single illustrative candidate stream* (seed 12345, budget 200): Q^2 is the outer predictive coefficient, RMSE the outer root-mean-square error, and “LO better” counts outer folds (of 18) in which the Loyola model had the strictly smaller absolute error. Panel (b) summarizes the 100-seed sweep (seeds 0–99): median ΔQ^2 with its [minimum, maximum], and the number of seeds in which $\Delta Q^2 > 0$, at budgets $B = 200$ and $B = 500$. All selection is repeated inside every outer fold; all values are full-precision results rounded once.

(a) single candidate stream: seed 12345, budget 200

Property	Q_{GM}^2	Q_{LO}^2	ΔQ^2	RMSE _{GM}	RMSE _{LO}	LO better
T_B	+0.613	+0.748	+0.136	3.700	2.983	10/18
ΔH_f	+0.855	+0.549	−0.306	0.463	0.816	4/18
ΔH_{vap}	+0.884	+0.971	+0.086	0.127	0.064	13/18
S	+0.906	+0.878	−0.029	1.363	1.558	7/18
ω	+0.979	+0.956	−0.022	0.005	0.008	5/18

(b) 100 candidate streams: seeds 0–99

Property	median ΔQ^2 [min, max]		seeds with $\Delta Q^2 > 0$	
	$B = 200$		$B = 200$	$B = 500$
T_B	+0.010	[−0.524, +0.373]	+0.066 [−0.307, +0.231]	53/100
ΔH_f	−0.120	[−0.408, +0.206]	−0.095 [−0.373, +0.118]	14/100
ΔH_{vap}	+0.069	[+0.034, +0.102]	+0.068 [+0.024, +0.102]	100/100
S	−0.012	[−0.140, +0.038]	−0.010 [−0.098, +0.031]	34/100
ω	−0.010	[−0.074, +0.016]	−0.006 [−0.052, +0.012]	26/100

lected the same (α, β) in all 18 folds there, while the three-parameter search selected widely varying triples (fold-to-fold standard deviation above 1.2 in both α and β). The selected γ values are interior to the search box (maximum $|\gamma| = 0.48$), but the parent’s selections are not: for ΔH_{vap} the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds. The consequences of this are examined in Section 6.2 of the main paper. On the one property where γ helps most stably, ΔH_{vap} , the identical triple $(-0.270, +0.271, +0.239)$ was selected in every fold. With $n = 18$ molecules these Q^2 differences carry substantial selection noise and we attach no inferential claim to them.

Because the comparison depends on a random candidate stream, the full nested design was repeated for one hundred seeds (0–99) at both budgets (200 and 500 candidates per model; 1000 configurations in all), together with a paired control in which the Loyola candidates are compared against the *same* (α, β) coordinates with γ set to zero, isolating the effect of γ from the independent candidate streams. Freeing γ improved nested predictive performance for ΔH_{vap} in every seed of this sweep (100/100 at both budgets, and 100/100 in the paired control; median $\Delta Q^2 = +0.069$ and $+0.068$, range $+0.024$ to $+0.102$) and *usually* degraded it for ΔH_f (86/100 and 76/100 seeds); T_B is close to an even split at budget 200 (53/100) but leans towards improvement at budget 500 (64/100), and S and ω usually degrade slightly. These counts describe optimizer variance over random candidate streams on one fixed set of 18 molecules; the molecules are never resampled, so the counts carry no inferential weight — in particular, 100/100 does not establish that γ improves ΔH_{vap} prediction for other molecules or data sets. The result is also sensitive to a single data cell: the ΔH_{vap} value of 2,2,3,3-tetramethylbutane, whose physical reference state is unresolved (Section S1). Repeating the complete nested comparison with that isomer excluded (17 molecules, same seeds, budgets and candidate streams), freeing γ improved ΔH_{vap} in 93/100 seeds at budget 200 and 96/100 at budget 500 (paired control 95/100 and 96/100; median $\Delta Q^2 = +0.046$ and $+0.051$, range -0.048 to $+0.098$), so the gain survives the exclusion but is no longer sign-invariant. These comparisons share the box $(-2, 2)$, which truncates the parent along the direction $\beta \approx -2\alpha$;

Section 6.2 and Table 5 of the main paper show that the vaporization-enthalpy gain shrinks to about zero once the parent is searched over a box that does not truncate it.

S4 Nonane replication test: protocol, exploratory transfer and molecule bootstrap

Two limitations of the octane analysis are its size (18 molecules) and its restriction to index models. Both were addressed under a protocol recorded in a git commit of the reproducibility package before any nonane value was collected: the identical nested design was applied to the 35 constitutional nonane isomers, with boiling points (35 values) and standard vaporization enthalpies (34 values; none listed for 3-ethyl-4-methylhexane) taken from the NIST WebBook [5] with every value traced to its archived source page. The two properties differ in evidential status. Twenty of the 35 boiling points are NIST averages of experimental determinations (13 quoted only to 1 K), five are single NIST determinations and ten are the most recent primary determination listed by NIST (Deviation D1 of the protocol); all are experimental measurements, so the T_B test is experimentally grounded. Of the vaporization enthalpies, only 4 of 34 are calorimetric determinations and one is a NIST average; 29 come from two compilations, the 1961 tabulation of Labauf, Greenshields and Rossini [4], which covers all 35 nonanes and 75 decanes and is not a set of individual calorimetric measurements, and a 1971 vapour-pressure handbook [8]. Because such tabulations may rest partly on structural-increment estimates, agreement with them measures consistency with the compiled values rather than independent experimental validation, and a degree-pair-based predictor may be favoured if the compiled values were themselves derived from structural increments. The pre-registered decision rule called the γ effect *replicated* if the Loyola model beat the parent in at least 80 of 100 seeds at both budgets with positive median ΔQ^2 . Four baselines were added, each fitted entirely within the outer training fold: the training-fold mean; the best of

the ten classical indices of Table 4 of the main paper; the irregularity index *IRLA*; and ridge regression on the full edge-degree-pair count vector, with its penalty chosen by inner leave-one-out. The resulting Q^2 values are in Table 6 of the main paper.

On the nonane replication set the pre-registered verdicts are: for ΔH_{vap} (compilation values) the γ effect met the pre-registered optimizer-stability criterion (*replicates*: LO better than GM in 95/100 and 98/100 seeds, median $\Delta Q^2 = +0.048$ and $+0.053$), and for T_B it is *mixed* (58/100 and 63/100; unchanged in direction under the pre-registered sensitivity analysis D4 of the protocol, which replaces the NIST average for 2,3,3,4-tetramethylpentane, 413 ± 6 K, by the median of all eight listed determinations including a 1989 handbook value and a 1939 outlier, 414.62 K: LO better in 60/100 and 66/100 seeds, median $\Delta Q^2 = +0.041$ and $+0.052$). The pre-registered protocol fixed the box of half-width 2, which, as Table 5 of the main paper shows, binds the parent; in a box that does not, the nonane gain disappears (median $\Delta Q^2 = -0.002$ under the hybrid control). The pre-registered verdict is reported as obtained, with this deviation recorded. The pattern behind the vaporization result is the same on both sets: models whose edge weight depends on imbalance only through q^2 inside the box of half-width 2 (the classical indices, including $GA = \sum_{uv} \sqrt{1 - q_{uv}^2}$, and GM) reach $Q^2 = 0.853\text{--}0.888$, while models with a term linear in q or with unrestricted counts (*IRLA*, ridge, $LO(0, 0, 1)$, LO) reach $Q^2 = 0.897\text{--}0.957$. This split holds only inside the narrow box: in the box of half-width 12 the parent alone reaches 0.942 and 0.892 (main paper, Table 5). The fixed point $LO(0, 0, 1)$, which involves no tuning at all, matches or beats the tuned Loyola model on both sets (0.957 and 0.903 against medians 0.953 and 0.900), and it is the index selected in every fold for ΔH_{vap} when admitted to the fixed-index comparison. It is essentially *IRLA*: $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2} IRLA(G) + O(\gamma^2)$, and on both sets the correlation between $LO(0, 0, 1)$ and *IRLA* is 0.996. The base-lines, however, also show the limit of the claim. Ridge regression on the full degree-pair count vector predicts the nonanes better than every index model, for both properties ($Q^2 = 0.927$ for ΔH_{vap} , 0.682 for T_B); on the octanes it matches the Loyola model on ΔH_{vap} and is weaker elsewhere.

This is consistent with Proposition 6 of the main paper and the remark after Proposition 7: the imbalance coordinate re-weights degree-pair information rather than adding to it, and its value is that one interpretable descriptor recovers most of what a model on all ten degree-pair counts achieves for the vaporization enthalpy, not that it outperforms such a model. Nor is the specific exponential form essential: the parameter-free aggregate *IRLA* comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900), so what matters for this property is a term linear in the imbalance, not the tuned exponential.

Finally, the nested design refits every model on the nonanes, which tests whether the *procedure* replicates, not whether an octane-trained model transfers. In an exploratory analysis (not pre-registered, and run in the box of half-width 2, so that it inherits the truncation of the parent), the parameters selected on all 18 octanes were frozen and applied to the nonanes. With the parameters frozen and only the linear calibration refitted, the Loyola model beat the parent for ΔH_{vap} in 100/100 seeds at both budgets (median $Q^2 = 0.910$ and 0.916 against 0.864 and 0.865) and for T_B in 65/100 and 82/100. With the octane calibration frozen as well, both models extrapolate poorly from C_8 to C_9 and the Loyola model is worse than the parent (median Q^2 between -2.22 and -0.72 for LO against 0.06 to 0.44 for GM). Because both comparisons inherit the truncated box, they do not bear on whether γ adds predictive information.

The seed counts above measure optimizer variance only. To indicate molecule-level uncertainty we also computed, post hoc, a paired bootstrap over molecules (2000 resamples) of the median-over-seeds ΔQ^2 , conditional on the fitted outer predictions (selection is not repeated within resamples, so the intervals do not account for uncertainty from refitting and model selection). The 95% intervals for LO minus GM on ΔH_{vap} are $[+0.014, +0.111]$ on the octanes but $[-0.006, +0.113]$ on the nonanes at budget 200 ($[-0.005, +0.119]$ at 500): the nonane gain met the pre-registered criterion, yet its interval narrowly includes zero. The interval for LO minus ridge on the nonanes, $[-0.101, +0.032]$, includes zero as well, and the T_B intervals are wide ($[-0.212, +0.222]$ for LO minus GM on the nonanes). The one octane interval that excludes zero in the other direc-

tion is that for ΔH_f , where freeing γ is worse ($[-0.281, -0.044]$). All these intervals refer to the box of half-width 2. Given Table 5 of the main paper, no γ effect remains to be established; the intervals are reported for completeness.

S5 Degeneracy across tree orders

Degeneracy measures an index’s ability to discriminate non-isomorphic graphs [3]; lower is better. On the 106 non-isomorphic trees of order 10 we use the distinct-value convention

$$\text{Degeneracy}(\%) = 100 \left(1 - \frac{k}{N} \right), \quad k = \#\{\text{distinct index values}\}, \quad N = 106,$$

i.e. the number $N - k$ of structures in excess of the distinct values, as a percentage of N (this is not the share of structures lacking a unique value, which is larger: for $LO(0,0,1)$ on this set 45 of the 106 trees, 42.5%, share their value with another tree). The pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ attain degeneracy 25.5% (79 distinct values), among the lowest of all compared indices and tied with R , GA , and AG ; indeed the 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the minimum degeneracy attainable by *any* degree-pair-based index on this set, and the pure- γ points attain this floor; the reason, and the order at which this stops, are given by Theorem 2 of the main paper; the first Zagreb index $LO(0,1,0)$ (83.0%) and the second Zagreb index $LO(1,0,0)$ (67.0%) are the most degenerate (Figure S2).

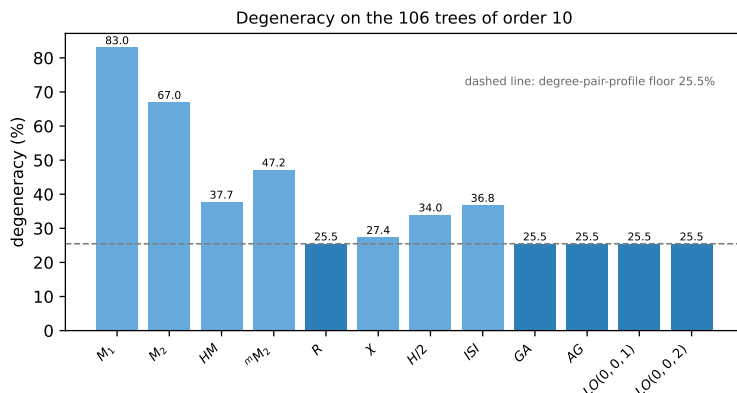


Figure S2. Degeneracy percentage of the compared indices on the 106 non-isomorphic trees of order 10. The dashed line marks the degree-pair-profile floor, $25.5\% = 100(1 - 79/106)$: the 106 trees realise only 79 distinct edge-degree-pair count vectors, so no vertex-degree-based index can fall below it.

To test whether this floor attainment is peculiar to order 10, the same computation was repeated for all non-isomorphic trees of orders 7 through 12 and, separately, for the molecular subsets with $\Delta \leq 4$ (Table S4). The pure- γ points attain the degree-pair-profile floor at *every* order in both sets. Honesty requires noting that the Randić index also attains the floor at every order in this range (while χ falls off the floor from order 10), so floor attainment distinguishes the pure- γ points from the most degenerate classical indices (M_1 , M_2) but not from R ; the observation is that the imbalance weights lose nothing, not that they are unique in this respect. Theorem 2 of the main paper turns this into an exact statement: the pure- γ points separate precisely the q -histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. Among the trees of order 13 the profiles $\{(1, 2)^1, (1, 3)^2, (1, 6)^5, (2, 3)^2, (2, 6)^1, (3, 3)^1\}$ and $\{(1, 3)^3, (1, 6)^5, (2, 2)^1, (2, 3)^2, (3, 6)^1\}$ (exponents are edge multiplicities) share the q -histogram $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$, and the pure- γ points realise 566 distinct values against 570 profiles; among molecular trees ($\Delta \leq 4$) the first failure occurs at order 16 (809 values against 810 profiles). The Randić index still attains the floor at order 13 (570 values) and falls below

it at order 14 (1095 values against 1100 profiles, where the pure- γ points give 1090).

Table S4. Discrimination across tree orders $n = 7$ –12: number of trees N , number of distinct edge-degree-pair profiles (the floor attainable by any vertex-degree-based index), and distinct-value counts for representative indices. “mol.” denotes the molecular subset $\Delta \leq 4$.

n	set	N	floor	M_1	M_2	R	$LO(0, 0, 1)$	$LO(0, 0, 2)$
7	all	11	11	7	9	11	11	11
7	mol.	9	9	5	7	9	9	9
8	all	23	21	9	17	21	21	21
8	mol.	18	16	6	13	16	16	16
9	all	47	40	13	22	40	40	40
9	mol.	35	28	7	16	28	28	28
10	all	106	79	18	35	79	79	79
10	mol.	75	49	8	21	49	49	49
11	all	235	152	21	43	152	152	152
11	mol.	159	83	9	25	83	83	83
12	all	551	294	27	57	294	294	294
12	mol.	355	137	10	28	137	137	137

Fixed fractional exponents (exploratory). A script in the repository (directory `structural`) evaluates the nine fixed points

$$(\alpha, \beta) \in \left\{ (0, \tfrac{1}{2}), (\tfrac{1}{2}, 0), (0, -\tfrac{1}{2}), (-\tfrac{1}{2}, 0), (\tfrac{1}{2}, -1), \right. \\ \left. (-\tfrac{1}{2}, 1), (\tfrac{1}{3}, 0), (0, \tfrac{1}{3}), (\tfrac{1}{6}, 0) \right\}$$

with $\gamma = 0$ and $\gamma = 1$. For discrimination on all trees of orders 12–16, every point off the ratio plane with $\gamma = 1$ reaches or nearly reaches the profile floor (e.g. order 16: 4069, 4067, 4069, 4068, 4069 and 4069 of 4069 for $(0, \tfrac{1}{2}), (\tfrac{1}{2}, 0), (0, -\tfrac{1}{2}), (-\tfrac{1}{2}, 0), (\tfrac{1}{3}, 0), (0, \tfrac{1}{3})$), whereas with $\gamma = 0$ the same points give 3475, 3940, 3476, 4007, 4009 and 3476; the two points on the ratio plane give 3996 for both values of γ . For prediction (fixed points, leave-one-out Q^2 , no tuning) $\gamma = 1$ beats $\gamma = 0$ in 52 of 81 point–dataset combinations, mostly for the vaporization enthalpy and decane boiling points, and loses mostly on the octane entropy and acentric factor; every fixed fractional point stays at $Q^2 \leq 0.21$ on the decanes. The full

output is committed with the script.

S6 Structure sensitivity: protocol and published control

The structure-sensitivity criterion for descriptor quality was introduced by Furtula, Gutman and Dehmer [2], with a fingerprint-based variant due to Rakić and Furtula [7]. We implement the published protocol on the 75 decane isomer trees Ψ : for each tree G , the similar-structure set $S(G) \subset \Psi$ consists of the non-isomorphic trees obtainable from G by relocating a single edge within the molecular class (graph edit distance two, $\Delta \leq 4$ preserved); then

$$SS(TI, G) = \frac{1}{|S(G)|} \sum_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

$$Abr(TI, G) = \max_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

and the class-level $SS(TI)$ and $Abr(TI)$ are the means over all 75 trees. On this class every $S(G)$ is nonempty (sizes 7 to 43), the neighbour relation is symmetric, and $Abr \geq SS$ holds throughout, as it must. As an external control, the implementation reproduces the published decane-class values of Barman and Das [1] for the seven indices common to both studies (M_1 , mM_2 , R , χ , H , GA , AG ; fourteen tabulated SS/Abr values): eleven agree exactly at the four displayed decimals and the remaining three differ by one unit in the fourth decimal (the cell-by-cell comparison is in the file `fgd_published_control.csv` of the reproducibility package; our values for all twelve compared indices are in Table 3 of the main paper). Under the published desideratum (SS large, Abr small [2]), the pure- γ point $LO(0, 0, 2)$ has a structure sensitivity equal to that of the hyper-Zagreb index (0.1257 against 0.1256) at a markedly smaller abruptness (0.274 against 0.313), and the sensitivity-to-abruptness ratios of the pure- γ points (0.4579 and 0.4565) are comparable to those of $H/2$, GA and R (0.448–0.451); in view of the redundancy analysis of Section 6.1 of the

main paper and Section S1, these third-decimal differences are not a ranking. Two qualifications limit this. First, SS is not a fixed property of the family: along the pure- γ line it grows with γ (0.031, 0.062, 0.126, 0.319 at $\gamma = 0.5, 1, 2, 5$) while the ratio peaks near $\gamma = 2$ and then falls (0.427 at $\gamma = 5$). Second, the parameter-free aggregate $IRLA$, which the pure- γ points approximate to first order in γ , has a higher ratio ($SS = 0.2343$, $Abr = 0.5066$, ratio 0.4625). The favourable sensitivity–abruptness balance is therefore a property of linear imbalance weighting rather than of the exponential form, and the SS margin over hyper-Zagreb is a tie, not a ranking.

S7 Lean 4 formalization: scope and encoding

The directory `formal/` of the repository contains a Lean 4 formalization (Lean v4.34.1, Mathlib v4.34.1) of the following statements of the main paper; `formal/verify.sh` builds it and checks that no `sorry`, `axiom` or `native_decide` occurs and that every theorem depends only on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

Encoding. Graphs are not formalized as Mathlib graphs; an edge set is a finite family of edges with a map to (positive real) degree pairs, or a multiset of degree pairs. Hypotheses such as “maximum degree Δ ” are stated as bounds on these pairs, so the link “these are the vertex degrees of a graph” is not formalized, except for the two order-13 trees below, which are realised by explicit parent arrays.

Formalized (inequalities and identities, with their hypotheses): the upper and lower (Δ, δ) bounds; the Cauchy–Schwarz interpolation bound; the ratio bound (and the bracket form); the convexity of $\theta \mapsto \log LO(G; \theta)$ for any nonempty edge set (via Hölder); the spanning statement of the independence lemma for every $\Delta \geq 3$ (via $\det W = \frac{1}{6} \ln 2 (8 \ln 3 - 13 \ln 2) \neq 0$, using $3^8 \neq 2^{13}$); the identity $1 - q^2 = 4xy/(x+y)^2$; and the order-13 profile pair with equal q -histograms and hence equal pure- γ values for every real γ , together with explicit 13-vertex trees realising both profiles.

Not formalized: the equality characterizations; the gradient and Hessian formulas and strict convexity; the ratio-plane theorem, the γ -crossing

bound and the identifiability proposition (which rely on the Lindemann–Weierstrass theorem, Laguerre’s rule and the theory of exponential families, not available in Mathlib); and all statistical and chemometric results; the closed forms, the $\gamma \rightarrow \pm\infty$ asymptotics, the ceiling and generic-discrimination propositions, and the identifiability consequence of Lemma 1 together with its $\Delta = 1, 2$ cases. A theorem-by-theorem map is in `formal/README.md`.

S8 Further directions

Building on the Loyola index, the following open problems indicate natural directions for further work:

1. Sharp two-sided bounds for $LO(G; \alpha, \beta, \gamma)$ via degree-based indices lying outside the family, such as the atom-bond-connectivity index ABC and the Sombor index.
2. Extremal-graph characterisations (not attempted in this paper) for several graph classes, including trees, unicyclic and bicyclic graphs.
3. Chemometric extension to additional chemical compound classes such as benzene derivatives, alkane homologues beyond the octanes, nonanes and decanes studied here and to compound-class-specific QSPR endpoints.

The first orders at which the pure- γ points, the Randić index and GA lose discrimination are summarised in Table 2 of the main paper.

S9 Closed forms, the log-convexity proof and termwise bounds

This section collects material moved from the main paper: closed-form values on standard graphs, the proof of the global log-convexity theorem (Theorem 1 of the main paper), and termwise bounds for the family.

Closed-form values on standard graphs

Proposition 1. *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

$$1. LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta} \text{ for } n \geq 2.$$

$$2. LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta} \text{ for } n \geq 3.$$

$$3. LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$$

$$4. LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right) \text{ for } n \geq 2.$$

$$5. LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right).$$

$$6. LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta} \text{ for } k \geq 1.$$

7. For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3, 3)$, so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$

8. For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2, 2)$, so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha (2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$

9. For an r -regular graph G with m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$.

Proof. Each case follows by enumerating the edge degree pairs and applying the definition of the main paper. In K_n , C_n , Q_k and r -regular graphs every edge has $d_u = d_v = r$, so $\text{imb} = 0$ and $(d_u d_v)^\alpha (d_u + d_v)^\beta = 2^\beta r^{2\alpha+\beta}$, which gives items (1), (2), (6) and (9). All $n-1$ edges of S_n have degree pair $(1, n-1)$ with $\text{imb} = (n-2)/n$ (item (4)); all pq edges of $K_{p,q}$ have pair (p, q) (item (5)); P_n , $n \geq 4$, has two pendant edges $(1, 2)$ and $n-3$ interior edges $(2, 2)$, and the cases $n = 2, 3$ are immediate (item (3)); items (7) and (8) follow by summing over the two edge-degree-pair classes of W_n and F_p . $\blacksquare$

Proof of the global log-convexity theorem

Proof of Theorem 1 of the main paper. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$, the (k, l) -entry of $\text{Cov}_\theta(z)$, which is positive semidefinite; convexity follows. The measure $w(\theta)$ has full support on E for every θ ,

so $\text{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace; this proves the strictness criterion. Since $q_{uv} \in [0, 1)$, $\mathbb{E}_\theta[q] \in [0, 1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r -regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ■

Termwise bounds

We give a bracket bound via the edge-contribution function ψ , bounds in (m, Δ, δ) for non-negative parameters, a Cauchy–Schwarz interpolation bound, a uniform ratio bound, and the $\gamma \rightarrow \pm\infty$ asymptotics; the $\gamma \neq 0$ bounds rest on bounding $e^{\gamma q_{uv}}$ edge by edge (for $\gamma \geq 0$) between 1 and $e^{\gamma(\Delta-1)/(\Delta+1)}$. Throughout, G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. Several of the inequalities have been machine-checked (Section S7).

Remark 1 (Attribution at $\gamma = 0$). At $\gamma = 0$ the family coincides with $M_{\alpha, \beta}$, for which inequalities of all the types developed here are already known: (Δ, δ, m) -bounds with regular/biregular equality and two-sided comparisons against M_1 , M_2 , R , χ , H , GA , AG and ISI [10], Jensen- and Hölder-type relations [9], and further inequalities [11]. The $\gamma = 0$ specializations below are therefore not claimed as new; the content of this section is their extension to $\gamma \neq 0$, where the imbalance factor $e^{\gamma q_{uv}}$ enters every edge weight, and the tracking of how γ deforms the equality cases (the $\gamma = 0$ comparisons with the Randić, sum-connectivity and Sombor indices are not repeated).

The summand in the defining sum of the main paper depends only on the unordered edge degree pair (d_u, d_v) . We isolate this dependence as follows:

Definition 1. For real parameters α, β, γ and positive reals x, y ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (\text{S1})$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 2. Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (\text{S2})$$

where, $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$, and the lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Every edge uv has its degree pair in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$, so $\psi_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$ on every edge; summing over the m edges and using $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives (S2). Each side of (S2) is the sum of m per-edge inequalities, so it is an equality if and only if every per-edge inequality is, i.e. if and only if every edge has a degree pair attaining $\psi_{\max}$ (upper side), respectively $\psi_{\min}$ (lower side), over $\mathcal{P}(\delta, \Delta)$. ■

Corollary 3. *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G without isolated vertices,*

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of G .

Proof. For each edge $e = uv$ write $\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) > 0$, $k \in \{1, 2\}$. By additivity of the exponents and the hypotheses on $(\alpha_k, \beta_k, \gamma_k)$, $\psi_e^{(1)} \psi_e^{(2)} = \psi(d_u, d_v; \alpha, \beta, \gamma)^2$ on every edge. The Cauchy–Schwarz inequality for $a_e = (\psi_e^{(1)})^{1/2}$ and $b_e = (\psi_e^{(2)})^{1/2}$ gives $\sum_e a_e b_e \leq (\sum_e a_e^2)^{1/2} (\sum_e b_e^2)^{1/2}$, which is the claim because $\sum_e a_e b_e = LO(G; \alpha, \beta, \gamma)$, $\sum_e a_e^2 = LO(G; \alpha_1, \beta_1, \gamma_1)$ and $\sum_e b_e^2 = LO(G; \alpha_2, \beta_2, \gamma_2)$. Equality holds if and only if $b_e = \lambda a_e$ for a constant $\lambda \geq 0$ and all e , i.e. if and only if $\psi_e^{(2)} / \psi_e^{(1)} = \lambda^2$ is the same on every edge. (The inequality itself is midpoint log-convexity of $\theta \mapsto LO(G; \theta)$, Theorem 1 of the main paper.) ■

Proposition 4. *Let G be a connected graph of size m and maximum degree Δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (\text{S3})$$

Proof. On every edge uv we have $1 \leq d_u, d_v \leq \Delta$, hence $d_u d_v \leq \Delta^2$ and $d_u + d_v \leq 2\Delta$; and the normalised imbalance $\text{imb}(x, y) = |x - y| / (x + y)$ is maximised over $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at $(1, \Delta)$, so $|d_u - d_v| / (d_u + d_v) \leq (\Delta - 1) / (\Delta + 1)$. Since $\alpha, \beta, \gamma \geq 0$, the maps $t \mapsto t^\alpha$, $t \mapsto t^\beta$ and $t \mapsto e^{\gamma t}$ are non-decreasing, so $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha} (2\Delta)^\beta \exp(\gamma(\Delta - 1) / (\Delta + 1))$; summing over the m edges yields (S3). For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for

connected G , $G \cong K_{1,\Delta}$); if $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ■

Proposition 5. *Let G be a connected graph of size m and minimum degree δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (\text{S4})$$

with equality if and only if G is δ -regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. On every edge $d_u, d_v \geq \delta$, so for $\alpha, \beta \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$ and $(d_u + d_v)^\beta \geq (2\delta)^\beta$, and for $\gamma \geq 0$ the exponential factor is at least 1 because the imbalance is non-negative. Multiplying and summing over the m edges yields (S4). If $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ equals the bound for every graph. Otherwise, equality requires equality in every active per-edge factor: if $\alpha > 0$ or $\beta > 0$ this forces $d_u = d_v = \delta$ on every edge; if $\alpha = \beta = 0$ and $\gamma > 0$ it forces $d_u = d_v$ on every edge, so the connected graph G is regular, hence δ -regular. Conversely, on a δ -regular graph every edge has degree pair (δ, δ) and equality holds. ■

The next proposition gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Proposition 6. *Let G be a simple graph with maximum degree at most Δ , and let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that appears in G . Define*

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in G . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G; \alpha, \beta, \gamma) \leq c_{\max}^h \cdot I_h(G), \quad (\text{S5})$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h .

Proof. Since $h(d_u, d_v) > 0$ and $c_{\min}^h \leq \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) \leq c_{\max}^h$ on every edge uv , multiplying by $h(d_u, d_v)$ and summing over $E(G)$, with $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$, gives (S5); each side is an equality if and only if every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition 6 is a *sharp graph-specific* comparison: its constants depend on the degree pairs that actually appear in G . A uniform version follows immediately by enlarging the range of the extrema.

Corollary 7 (Uniform Δ -bounded version). *With the hypotheses of Proposition 6, but with h strictly positive on all of $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ and the extrema taken over P_Δ , inequality (S5) holds simultaneously for every graph of maximum degree at most Δ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ ; the uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

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